



Fig. 4: Typical tank firing



Fig. 5: American AGM-114 Hellfire laser-guided missile on predator drone. Missile's front tip is the eye (Photograph credit: US Department of Defense, through Wikipedia)



Fig. 6: American BGM-71 tube-launched optically-tracked wire-commanded data link guided missile with the wire trailing behind the missile (Photograph credit: US Army, through Wikipedia)

of an engineering process than a firing process.

During the First World War, in the Battle of the Somme, a strange-looking beastly armoured metallic vehicle making 'clang clink clang' noises crawled into the battlefield like a caterpillar. It was a British invention that was conspicuously named as 'tank' to conceal its real purpose of crossing trenches. As the trench warfare of the WW-I went out of favour of the militaries, tanks got a new job. That job was to be the spearhead of an advancing force.

Germans exploited this aspect through their blitzkrieg campaigns in WW-II and tanks came to prominence.

Unlike the artillery guns, tanks see their targets and shatter them. During an advance, typically, the tank will put its crosshairs on anything that annoys the advance and blow it into smithereens. If a bunker is annoying the infantry the tank will put it in its crosshairs and shatter it, in what is called 'direct fire support.' The shells fired by tanks exit the gun in a flat trajectory at a speed of around 1525 m/s (5000 ft/s) whereas the shells fired from SPGs exit the gun in a parabolic trajectory at a speed of around 305 m/s (1000 ft/s).

The basic differences between a tank and an SPG make them to complement each other. In an armoured thrust deep inside the enemy territory, the tanks form

frontline and proceed shattering what they see. The SPGs follow the frontline at a distance of 20 km, giving indirect fire support as and when required. With this background let us embark on a virtual tour to see the electronic systems present in these big guns – the defensive, offensive and command & control systems.

Crew protection systems

Losing a well-trained crew is more unacceptable for an army than losing a tank or self-propelled gun. So these big

guns have armour around them. As the self-propelled guns are not operated on the frontlines, these have minimum armour. This armour can protect the crew only from bullets and splinters, but not from an enemy tank's shell. On the other hand, as the tanks form the spearhead of an invasion force, they require enormous protection that is usually provided by its mightily heavy armour.

The tank's heavy armour, through its bulk and tensile strength, protects the crew to a greater extent even from the enemy shells. This armour itself became an iconic representation of tanks within a few years of tanks' introduction into battlefield. Army units centred on tanks got named as armoured divisions, armoured corps, etc, and now form the core of the offensive formations all over the world.

Introduction of active protection systems (APS) for crew has made the armour just the last line of defence. The APS protect the tank from anti-tank missiles and rockets. Hard-kill APS (HK-APS) destroy the projectiles while soft-kill APS (SK-APS) spoofs and diverts them.

Soft-kill active protection systems

SK-APS are used against laser-homing and wire-guided anti-tank missiles. Laser-homing missiles look for laser reflections from the targeted tank to home in and attack. For this, a laser designator is focussed on the targeted tank by another entity. The designator continuously emits pulse-coded IR laser beam which gets meagrely reflected by the tank. After the missile is launched, while it is flying, it looks for these meagre reflections from the targeted tank, spots them, and flies towards the reflection (tank) to destroy it.

A wire-guided anti-tank missile is launched from a command-launcher console. The missile uncoils a thin wire while flying, throughout its range. The command-launcher console sends guidance signal to the missile through this wire. The operator just keeps the target on the crosshairs of the command-launcher console. A computer